PEARSON EDEXCEL INTERNATIONAL GCSE · FURTHER PURE MATHEMATICS · 4PM1/01

A* Examination Guide

Built from 16 past papers and their mark schemes, January 2019 to June 2026

This guide is built from the 4PM1 Paper 1 question papers and mark schemes themselves. Methods are set out the way the mark schemes award M and A marks, and every warning comes from a note that examiners applied: “A0”, “withhold”, “do not allow”, “answers with no working score no marks”. Use the formula booklet for what it gives you, and learn the rest.

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Papers analysed: Jan 2019 (legacy 4PM0), Jun 2019, Jan 2020, Nov 2020, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 24).

1 How Paper 1 is built

Format: 2 hours, 100 marks, 10 or 11 questions: about 1.2 minutes per mark. Calculators are allowed, but the front page warns “Without sufficient working, correct answers may be awarded no marks”, and many parts say “use algebra”, “use calculus” or “show your working clearly”. The formula sheet gives mensuration for spheres and cones, AP and GP sums, the binomial series, the quotient rule, the cosine rule, sin/cos/tan of (A ± B) and the change-of-base rule for logs. Everything else must be learnt.

Topic	Papers (of 16)	What is asked most
Differentiation and its applications	16	product/quotient/chain, normals, stationary points, optimisation
Integration: areas and volumes	15	area between curve and line, volume of revolution, exact answers
Series (AP, GP, Σ)	13	find a, d or r; sum to infinity; least n using logarithms
Completing the square and the discriminant	11	$a(x + b)^2 + c$, min/max, set of values of k
Binomial expansion	10	$(1 + x)^n$, rewrite $(a + bx)^n$, approximate a root
Logarithms	10	solve log equations with different bases; simultaneous log equations
Coordinate geometry	9	perpendicular lines, ratio points, areas of quadrilaterals
Rates of change and small changes	8	spheres, cones, hemispheres; $\delta y \approx (dy/dx)\delta x$
Factor and remainder theorems	8	find constants, then factorise and solve a cubic
Triangles and 3D trigonometry	13	sine/cosine rule, area, pyramids and prisms, angle between planes
Trig identities and equations	7	double angles from the formula sheet; solve in a range
Vectors	7	ratio points, collinearity, λ and μ, area ratios
Roots of quadratics (α, β)	7	$\alpha^3 + \beta^3$, new equations with integer coefficients

Topic	Papers (of 16)	What is asked most
Linear inequalities and graphs	12	shade a region, optimise P = ax + by; solve by drawing a line
Kinematics	5	v → a by differentiation; v → s by integration; distance

The recurring long questions

- **Optimisation (6–8 marks):** open box (Jan 19), prism (Jan 20), cylinder (Jan 22), badge (Jan 23), cuboid (Jun 23), quarter-circle shape (Nov 24): form S or P, differentiate, set to zero, justify with d^2y/dx^2.
- **Area or volume by algebraic integration (6–9):** almost every paper, often the last question.
- **Coordinate geometry chain (8–12):** line → perpendicular → point with given distance and gradient → area.
- **Graph work (6–8):** a table, a curve, then "by drawing a suitable straight line … estimate the roots of …".

2 TWENTY MARK-SCHEME RULES

1. **Correct answers with no working score no marks** where the question says "use algebra", "use calculus" or "show your working" (e.g. "Correct solution seen with no working scores B0M0A0M0A0A0", Nov 21).
2. **Calculator solutions of a quadratic:** if your quadratic is wrong and no method is shown, the method mark is lost. Factorise or use the formula on the page.
3. **"Show that" and given answers:** every step, no missing brackets, no incorrect working. "This is a show question so every step must be seen" (Jan 20).
4. **"Hence":** use the previous result. Starting afresh may score nothing.
5. **Integration must be shown:** "Solutions where the integration step has not been shown will score M0 M0 A0" (Jun 24); "No evidence of algebraic integration – no marks".
6. **Substitute the limits visibly** when the final answer is wrong; the method mark needs to see both limits.
7. **Exact answers stay exact:** using rounded angles or sines in an "exact" answer is A0 (Nov 20).
8. **Radians for calculus with trig and for sectors:** a substitution in degrees is not accepted (Jun 21).
9. **Degrees when the question is in degrees,** to the accuracy asked: "Do not accept awrt … rounding must be to one decimal place" (Jun 23).
10. **Extra solutions inside the range lose the final A mark;** values outside the range are ignored.
11. **Do not discard valid roots,** and reject invalid ones with a reason (negative lengths, log of a negative).
12. **Asymptotes are equations:** "x = −5" not "−5" (Jun 21). Marks are not given retrospectively from a sketch.
13. **Sketches of rational curves:** both branches approach the asymptotes without crossing or bending back.
14. **Small changes:** δx is the actual change (0.2% of 2.5 = 0.005), not 0.2 or 0.2% (Jun 23).
15. **Least n with logarithms:** show the inequality, the log step and the rounding up to an integer.
16. **Binomial coefficients as exact fractions in lowest terms** when asked; ignore higher powers.
17. **Brackets:** "Do not allow missing brackets unless recovered later – this is a general point of marking."
18. **Equations in the requested form:** ax + by + c = 0 with integers; "P = …" not "Perimeter = …" (Jan 23).
19. **Graph estimates:** draw the straight line you use and read to one decimal place; joining points with straight segments loses the curve mark.
20. **Accuracy:** 3 s.f. unless told; the mark scheme often quotes the calculator value, so do not round early.

3 QUADRATICS, ROOTS AND POLYNOMIALS

COMPLETING THE SQUARE AND THE DISCRIMINANT

$ax^2 + bx + c = a(x + b/(2a))^2 + c - b^2/(4a)$. The turning point is $(-b/(2a), c - b^2/(4a))$; the line of symmetry is $x = -b/(2a)$.

Jun 26: $-3x^2 + 12x - 2 = -3(x - 2)^2 + 10$: maximum 10, line of symmetry $x = 2$.

Discriminant: $b^2 - 4ac > 0$ two distinct real roots; $= 0$ equal roots; < 0 no real roots. For a "set of values of k", solve the quadratic inequality and give both regions correctly ($k < \ldots$ or $k > \ldots$).

ROOTS A AND B

For $ax^2 + bx + c = 0$: $\alpha + \beta = -b/a$, $\alpha\beta = c/a$. Learn $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$, $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$, $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$.

Nov 21: $4x^2 - 3x - 8 = 0$, new roots $1/\alpha$ and $1/\beta$: sum $= (\alpha + \beta)/\alpha\beta = -3/8$, product $= -1/2 \Rightarrow 8x^2 + 3x - 4 = 0$.

New equation: $x^2 - (\text{sum})x + (\text{product}) = 0$, then multiply to integer coefficients.

FACTOR AND REMAINDER THEOREMS

$(x - a)$ is a factor $\Leftrightarrow f(a) = 0$; the remainder on dividing by $(x - a)$ is $f(a)$. Two conditions $\rightarrow$ two simultaneous equations for the constants.

Jan 20: $6x^3 - 31x^2 + 25x + 12 = (x - 4)(3x + 1)(2x - 3) = 0 \Rightarrow x = 4, -1/3, 3/2$. Show the division or comparison of coefficients.

- Linear-and-quadratic inequalities (Jun 26): solve each, then combine on a number line.
- A line meeting a curve at exactly one point (Jan 23): substitute and set the discriminant = 0.
- Surds (Jan 22, Jun 22): rationalise by multiplying by the conjugate; show every line "without a calculator".

4 LOGARITHMS

LAWS (CHANGE OF BASE IS ON THE FORMULA SHEET)

$\log_a x + \log_a y = \log_a xy$; $\log_a x - \log_a y = \log_a(x/y)$; $k \log_a x = \log_a x^k$; $\log_a x = \log_b x/\log_b a$. Useful: $\log_{a^{-1}} x^2 = \log_a x$, e.g. $\log_{49} x^2 = \log_7 x$.

Method: convert every term to one base, combine into a single log on each side, remove the logs, solve, and check every root in the original (arguments must be positive).

WORKED EXAMPLE · JAN 20

$\log_7(8x^2 - 6x + 3) - \log_{49} x^2 = 3 \log_7 2 \Rightarrow \log_7[(8x^2 - 6x + 3)/x] = \log_7 8 \Rightarrow 8x^2 - 14x + 3 = 0 \Rightarrow (4x - 1)(2x - 3) = 0 \Rightarrow x = 1/4$ or $3/2$ (both valid).

- Simultaneous log equations (Nov 23): rewrite each as a power equation, e.g. $\log_2 x = 3 \Rightarrow x = 8$, then substitute.
- Exponential graph questions (Jun 22, Jan 23, Nov 25, Jun 26): rearrange the given log equation to match the plotted curve, then draw the straight line that represents the other side.

5 SERIES

FORMULAE AND FACTS

AP: $u_n = a + (n - 1)d$; $S_n = (n/2)[2a + (n - 1)d]$. GP: $u_n = ar^{n-1}$; $S_n = a(1 - r^n)/(1 - r)$; $S_\infty = a/(1 - r)$, $|r| < 1$. Σ notation: $\Sigma_{r=1}^{n}(5r - 1)$ is an AP with $a = 4$, $d = 5$.

Σ from $r = 10$ to $n = (\Sigma$ from 1 to $n) - (\Sigma$ from 1 to 9).

WORKED EXAMPLE · LEAST N (JUN 24)

$a + ar = 400$, $ar + ar^2 = 100 \Rightarrow r = 1/4$, $a = 320$, $S_\infty = 1280/3$.

$S_n > 426.6 \Rightarrow (1280/3)(1 - 0.25^n) > 426.6 \Rightarrow 0.25^n < 1.5625 \times 10\text{^\{-4\}} \Rightarrow n > \ln(1.5625 \times 10^{-4})/\ln 0.25 = 6.32$ $\Rightarrow n = 7$. The inequality reverses when dividing by $\ln 0.25 < 0$.

- Pairs of conditions ($t_2 + t_9 = 0$, $t_4 + t_6 + t_{10} = 14$) give simultaneous equations in a and d.
- Ratios of sums ($S_{2n} = 3S_n$, Jan 20) and relations such as $2S_n = 5u_n + 10$ (Nov 24): write both sides in n, solve the quadratic and reject non-integer or negative n.
- An AP of logs (Jun 21): $u_n = (n + 1)\ln 4 = 2(n + 1)\ln 2$ is an AP with $d = \ln 4$.

6 BINOMIAL EXPANSION

METHOD

$(1 + x)^n = 1 + nx + n(n - 1)x^2/2! + n(n - 1)(n - 2)x^3/3! + \ldots$, valid for $|x| < 1$ (formula sheet). For $(a + bx)^n$ take out a^n: $(a + bx)^n = a^n(1 + bx/a)^n$, valid for $|x| < |a/b|$.

Jan 20: $(1 - 4x)^{½} = 1 - 2x - 2x^2 - 4x^3$. With $x = 0.06$, $\sqrt{0.76} \approx 0.871936$, so $\sqrt{19} = 5\sqrt{0.76} \approx 4.360$.

- Choose the substitution so the bracket gives the required root, and the x is inside the validity range.
- Products such as $(2 - x^2)(1 - 2x)^{-½}$ (Jan 19): expand the series, then multiply and keep terms to x^3.
- Integrating the series to estimate an area (Jan 19, Jun 22, Nov 23): integrate term by term with the given limits.
- Unknowns a, n (Jun 21) or a, b (Nov 23, Nov 24): equate given coefficients and solve.

7 COORDINATE GEOMETRY

TOOLKIT

Gradient $m = (y_2 - y_1)/(x_2 - x_1)$; perpendicular gradients multiply to -1; distance $\sqrt{[(\Delta x)^2 + (\Delta y)^2]}$; midpoint $((x_1 + x_2)/2, (y_1 + y_2)/2)$; point dividing AB in ratio $m : n = A + (m/(m + n))(B - A)$.

Area of a polygon: split into triangles, or use the shoelace formula $½|\Sigma(x_i y_{i+1} - x_{i+1} y_i)|$ with vertices in order.

WORKED EXAMPLE · JAN 23

A(−5, 3), B(4, 0), C(−1, 5). Gradient AB = −1/3, so l through C has gradient 3: $3x - y + 8 = 0$. D(−2, 2) lies on both lines. The midpoint of AB is (−0.5, 1.5) ≠ D, so l is not the perpendicular bisector of AB.

- Finding (m, n) from a distance and a gradient (Jan 19, Jan 22, Nov 25): form two equations, substitute, solve the quadratic, and use the stated condition ($m > 0$, $n < 0$) to choose.
- Area of a triangle with one vertex on a line (Jun 24): ½ × base × perpendicular height, or the shoelace formula.

8 GRAPHS, REGIONS AND ASYMPTOTES

LINEAR PROGRAMMING (JAN 20, NOV 20, JAN 22, NOV 24, JUN 25, NOV 25)

Draw each line accurately across the grid; shade or label the region R exactly as the question asks.

For $P = ax + by$, evaluate P at each vertex of R (read from the graph or solved algebraically) and choose the least or greatest. Show which vertex gives it.

SOLVING EQUATIONS GRAPHICALLY

Rearrange the equation so one side is the plotted curve; the other side is the straight line to draw. Read the x-coordinates of the intersections to one decimal place and show the line on the grid.

Example: with $y = 1 + 3e^{-x}$ plotted, $x = e^{-x} \Leftrightarrow 1 + 3e^{-x} = 1 + 3x$, so draw $y = 1 + 3x$ (Jun 22).

RATIONAL CURVES Y = (AX + B)/(CX + D) (JAN 19, JUN 21, JAN 23, JUN 24)

Vertical asymptote $x = -d/c$; horizontal asymptote $y = a/c$. Intercepts: x-axis where $ax + b = 0$; y-axis at $y = b/d$.

Sketch both branches approaching the asymptotes, label equations and intercepts. Gradient always positive or negative: show the sign of $dy/dx = (ad - bc)/(cx + d)^2$.

A line with no intersections (Jun 24): substitute, form a quadratic, and require the discriminant < 0.

9 TRIGONOMETRY AND 3D

RADIANS AND SECTORS

Arc length $s = r\theta$; sector area $\frac{1}{2}r^2\theta$; segment area $\frac{1}{2}r^2(\theta - \sin\theta)$. θ must be in radians.

Nov 21: $\frac{1}{2}r^2\theta = 16.8$ and $2r + r\theta = 16.4 \Rightarrow r^2 - 8.2r + 16.8 = 0 \Rightarrow r = 4$ (integer), $\theta = 2.1$.

TRIANGLES

Sine rule $a/\sin A = b/\sin B$; cosine rule $a^2 = b^2 + c^2 - 2bc\cos A$ (on the sheet); area $\frac{1}{2}ab\sin C$.

Ambiguous case: if the angle is obtuse (Nov 24) take $180° -$ the calculator value. A quadratic from the cosine rule (Nov 25: $x^2 - 12x - 81 = 0$) gives one positive length.

3D: PYRAMIDS AND PRISMS (NOV 20, NOV 23, NOV 24, JUN 25, JUN 26)

Angle between a line and a plane: the angle between the line and its projection on the plane.

Angle between two planes: find a line in each plane perpendicular to their common edge (usually from the vertex to the midpoint of a base edge), then use the right-angled triangle or the cosine rule.

Obtuse angle between faces (Jun 25): use the cosine rule with perpendiculars from the same point on the edge.

IDENTITIES AND EQUATIONS

From the sheet: $\sin(A \pm B)$, $\cos(A \pm B)$, $\tan(A \pm B)$. Derive $\sin 2A = 2\sin A\cos A$, $\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$, $\tan 2A = 2\tan A/(1 - \tan^2 A)$ when asked "using a formula on page 2".

Jan 20: $3\cos^2\theta - 3\sin^2\theta + \sin\theta + 12 = 0 \Rightarrow 6\sin^2\theta - \sin\theta - 15 = 0 \Rightarrow \sin\theta = 5/3$ or $-3/2$, both outside $[-1, 1]$, so no solutions. Saying "undefined" without $|\sin\theta| \le 1$ was not accepted.

Solve for $2x - 75$ etc. by widening the range first; list every solution in range and no others.

10 VECTORS

METHOD (JAN 20, NOV 20, JUN 21, NOV 21, JUN 22, JAN 23, JUN 23)

Write every vector in terms of the two base vectors (a, b). For a point dividing PQ in ratio m : n: $OX = OP + (m/(m + n))PQ$.

Intersection: express OX two ways ($OX = \lambda OB$ and $OX = OM + \mu MN$), equate coefficients of a and b, and solve for λ and μ.

Collinear: show one vector is a scalar multiple of the other and that they share a point.

Area ratios: triangles with the same height have areas in the ratio of their bases.

- Unit vector parallel to v: v/|v| (Jun 23). Magnitude $\sqrt{(x^2 + y^2)}$.
- Regular hexagon (Nov 21): opposite sides are equal vectors; OB = a + b, BC = b − a.

11 DIFFERENTIATION AND ITS APPLICATIONS

y	dy/dx	y	dy/dx
x^n	nx^{n-1}	e^{kx}	ke^{kx}
sin kx	k cos kx	ln x	1/x
cos kx	−k sin kx	uv	u′v + uv′
f(g(x))	f′(g(x)) g′(x)	u/v	$(u'v - uv')/v^2$ (on the sheet)

WORKED EXAMPLES

Jun 21 (normal): $y = (x^3 - 2x)e^{1-x}$ at (1, −1): $dy/dx = (3x^2 - 2)e^{1-x} - (x^3 - 2x)e^{1-x} = 2$. Normal gradient −½: x + 2y + 1 = 0.

Nov 20 (implicit-style): $2xy + 5y = e^x \Rightarrow y = e^x/(2x + 5)$; quotient rule gives dy/dx = y(2x + 3)/(2x + 5).

Stationary points: dy/dx = 0; nature from d^2y/dx^2 (> 0 minimum, < 0 maximum).

OPTIMISATION (6–8 MARKS)

Use the constraint to eliminate a variable, giving S (or V, P) in one variable; differentiate; set to 0; justify with d^2S/dx^2 at that value; substitute back.

Jan 22: $S = 2\pi r^2 + 180\pi/r \Rightarrow dS/dr = 4\pi r - 180\pi/r^2 = 0 \Rightarrow r^3 = 45$, r = 3.56; $d^2S/dr^2 = 4\pi + 360\pi/r^3 > 0$, so minimum; S = 238 cm^2.

RATES AND SMALL CHANGES

Connected rates: dA/dt = (dA/dr)(dr/dt). **Nov 21:** dV/dt = 3, r = 10: $dr/dt = 3/(4\pi r^2) = 3/(400\pi)$; $dS/dt = 8\pi r \times dr/dt = 0.6$ cm^2/s.

Small changes: $\delta y \approx (dy/dx)\delta x$. A 2% increase in x means $\delta x = 0.02x$; percentage change in $y = (\delta y/y) \times 100$.

KINEMATICS

a = dv/dt; s = ∫v dt (+ c from the initial position). At rest: v = 0. Distance: integrate |v|, splitting at the times when v = 0.

Jun 25: $v = 3t^2 - 9t + 7$: discriminant −3 < 0 so never at rest; a(4) = 15 m/s^2; distance $0 \rightarrow 5 = [t^3 - 4.5t^2 + 7t] = 47.5$ m.

Jun 23: $v = 2(t - 3)(t - 5)$: at rest at t = 3 and 5; distance $MN = |\int_3^5 v\ dt| = 8/3$ m.

12 INTEGRATION: AREAS, VOLUMES, KINEMATICS

RESULTS

$\int x^n\ dx = x^{n+1}/(n + 1) + c$ $(n \neq -1)$; $\int e^{kx}\ dx = e^{kx}/k$; $\int \sin kx\ dx = -\cos(kx)/k$; $\int \cos kx\ dx = \sin(kx)/k$; $\int 1/x\ dx = \ln x$.

Area under a curve: ∫y dx; regions below the axis give negative values, so integrate each part separately and add magnitudes. Area between curve and line: ∫(upper − lower) dx between intersections.

Volume about the x-axis: $\pi\int y^2\ dx$; about the y-axis: $\pi\int x^2\ dy$.

WORKED EXAMPLES

Jun 26: $f(x) = x^3 - 2x^2 - 9x + 18 = (x - 3)(x - 2)(x + 3)$. Region from the y-axis to C(3, 0): $\int_0^2 f\,dx - \int_2^3 f\,dx = 211/12$.

Jun 25: $y = \sqrt{(4x - 8)}$, rotated from x = 2 to x = b: $\pi\int(4x - 8)\,dx = 2\pi(b - 2)^2 = 50\pi \Rightarrow b = 7$.

Jan 22 (reverse): given $d/dx[e^{4x}(8x^2 - 4x + 1)] = kx^2e^{4x}$, use it to integrate $y^2 = 9x^2e^{4x}$ for the volume.

- Find limits from intersections (Jan 20, Jan 23, Jun 23) and show the algebra.
- Composite regions (Nov 24, Nov 23): volume of the curve part ± a cone from a straight line ($V = \frac{1}{3}\pi r^2 h$).
- Equal-area conditions (Jun 22): set the two integrals equal and solve the resulting polynomial.

13 FINAL A* CHECKLIST

1. Read the instruction words: “use algebra”, “use calculus”, “show your working clearly”, “exact”, “hence”.
2. Quadratics: method on the page (factorise or formula), not a calculator answer alone.
3. Logs: one base, one log each side, solve, check arguments are positive.
4. Series: least n with logs, flipping the inequality for logs of numbers below 1; round up.
5. Binomial: take out a^n, coefficients as fractions, state validity, show the substitution.
6. Trig: radians for sectors and calculus; degrees to the stated accuracy; all solutions in range only.
7. Calculus: chain/product/quotient correct way round; justify max/min with the second derivative.
8. Integration: show the integrated expression and both limits; separate regions below the x-axis.
9. Graphs: draw the extra straight line; read values to one decimal place; label region R.
10. Leave 10 minutes to recheck arithmetic, signs and the final accuracy.